

LAPORAN BAB3

Praktikum Devops

Docker Network, Volume, Bind Mount, tmpfs, dan Compose



Disusun Oleh:

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PROGRAM REKOGNISI PEMBELAJARAN LAMPAU (RPL)

JURUSAN TEKNIK INFORMATIKA

POLITEKNIK ELEKTRONIKA NEGERI SURABAYA

SURABAYA

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1 Tujuan Praktikum

- Membangun struktur direktori kerja terstandar untuk penyimpanan DevSecOps.
- Mencatat dan memverifikasi versi komponen serta engine yang digunakan untuk memastikan keterulangan (reproducibility) pengujian.
- Memeriksa mekanisme keamanan yang aktif pada host/daemon Docker melalui SecurityOptions.
- Menyusun landasan dokumentasi evidence dan *threat statement* awal.

2 Langkah Praktikum

2.1 User-defined bridge network

1. docker network create --driver bridge --subnet 172.20.0.0/16 lab-net

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker network create --driver  
bridge --subnet 172.20.0.0/16 lab-net  
bd48ee0fd387f9ae6a59ac9d8ae73c446d9759f5a1a3c846426e6227335bfe2b  
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

2. docker run -d --name server-a --network lab-net nginx:alpine

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker run -d --name server-a  
--network lab-net nginx:alpine  
Unable to find image 'nginx:alpine' locally  
alpine: Pulling from library/nginx  
55afa1ecc21d: Pull complete  
58c524ea09ce: Pull complete  
850bf2dcecff: Pull complete  
51900e10fb9c: Pull complete  
8f924cf5086c: Pull complete  
bc98d7675616: Pull complete  
6636b9fc203c: Pull complete  
af7dd138f459: Pull complete  
a32eccfe9457: Download complete  
10d8afe205cd: Download complete  
Digest: sha256:72ba65eb42c10344912a84ff42408db7d34f2feb642204570ab8fc5fffd29f  
1d3  
Status: Downloaded newer image for nginx:alpine  
8bf8fda98a7637b3764b4118424068fdd147f5519fd7b5e7662b9ef118af1db3  
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

3. docker run -d --name server-b --network lab-net nginx:alpine

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker run -d --name server-b  
--network lab-net nginx:alpine  
d32493f6e2982b6ee9858122a375ba4b06d5f7ecdd3c16c391f49898f6e6eab4  
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

4. docker exec server-a ping -c 3 server-b

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker exec server-a ping -c 3
server-b
PING server-b (172.20.0.3): 56 data bytes
64 bytes from 172.20.0.3: seq=0 ttl=64 time=0.313 ms
64 bytes from 172.20.0.3: seq=1 ttl=64 time=0.104 ms
64 bytes from 172.20.0.3: seq=2 ttl=64 time=0.196 ms

--- server-b ping statistics ---
3 packets transmitted, 3 packets received, 0% packet loss
round-trip min/avg/max = 0.104/0.204/0.313 ms
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

5. docker rm -f server-a server-b

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker rm -f server-a server-b
server-a
server-b
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

2.2 Volume backup dan restore

1. docker volume create data-vol

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker volume create data-vol
data-vol
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

2. docker run -d --name writer -v data-vol:/app/data alpine:3.20 sh -c "while true; do date >> /app/data/log.txt; sleep 5; done"

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker run -d --name writer -v
data-vol:/app/data alpine:3.20 sh -c "while true; do date >> /app/data/lo
g.txt; sleep 5; done"
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
25f1d6b1951a: Pull complete
3a030ca3f633: Download complete
d4050d56ebf2: Download complete
Digest: sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b
6bc
Status: Downloaded newer image for alpine:3.20
f7d0d4f3782a3798b12c4d4a5de4e3e3723329c9a46965ca84ede991577719f3
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

3. sleep 15

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ sleep 15
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

4. docker rm -f writer

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker rm -f writer
writer
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

5. docker run --rm -v data-vol:/data alpine:3.20 cat /data/log.txt

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker run --rm -v data-vol:/d
ata alpine:3.20 cat /data/log.txt
Fri Sep 11 16:03:17 UTC 2026
Fri Sep 11 16:03:22 UTC 2026
Fri Sep 11 16:03:27 UTC 2026
Fri Sep 11 16:03:32 UTC 2026
Fri Sep 11 16:03:37 UTC 2026
Fri Sep 11 16:03:42 UTC 2026
Fri Sep 11 16:03:47 UTC 2026
Fri Sep 11 16:03:52 UTC 2026
Fri Sep 11 16:03:57 UTC 2026
Fri Sep 11 16:04:02 UTC 2026
Fri Sep 11 16:04:07 UTC 2026
Fri Sep 11 16:04:12 UTC 2026
Fri Sep 11 16:04:17 UTC 2026
Fri Sep 11 16:04:22 UTC 2026
Fri Sep 11 16:04:27 UTC 2026
Fri Sep 11 16:04:32 UTC 2026
Fri Sep 11 16:04:37 UTC 2026
Fri Sep 11 16:04:42 UTC 2026
Fri Sep 11 16:04:47 UTC 2026
Fri Sep 11 16:04:52 UTC 2026
Fri Sep 11 16:04:57 UTC 2026
Fri Sep 11 16:05:02 UTC 2026
Fri Sep 11 16:05:07 UTC 2026
Fri Sep 11 16:05:12 UTC 2026
Fri Sep 11 16:05:17 UTC 2026
Fri Sep 11 16:05:22 UTC 2026
Fri Sep 11 16:05:27 UTC 2026
Fri Sep 11 16:05:32 UTC 2026
Fri Sep 11 16:05:37 UTC 2026
Fri Sep 11 16:05:42 UTC 2026
Fri Sep 11 16:05:47 UTC 2026
Fri Sep 11 16:05:52 UTC 2026
Fri Sep 11 16:05:57 UTC 2026
Fri Sep 11 16:06:02 UTC 2026
Fri Sep 11 16:06:07 UTC 2026
Fri Sep 11 16:06:12 UTC 2026
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

6. `docker run --rm -v data-vol:/source:ro -v $(pwd):/backup alpine:3.20 tar czf /backup/data-vol-backup.tar.gz -C /source .`

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker run --rm -v data-vol:/s
ource:ro -v $(pwd):/backup alpine:3.20 tar czf /backup/data-vol-backup.tar
.gz -C /source .
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

2.3 Compose multi-container Nginx-Flask-PostgreSQL

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ mkdir -p ~/docker-lab/bab-3/ap
p ~/docker-lab/bab-3/html
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ ls
app data-vol-backup.tar.gz html
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

```

app data-vol-backup.tar.gz html
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ cat << 'EOF' > ./html/index.html
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <title>Lab Docker Compose 3-Tier</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f6f9; }
    .card { background: white; padding: 25px; border-radius: 8px; box-shadow: 0 2px 5px rgba(0,0,0,0.1); }
    h1 { color: #1a365d; }
  </style>
</head>
<body>
  <div class="card">
    <h1>Selamat Datang di Lab Docker Bab 3!</h1>
    <p>Layanan Nginx Web Reverse Proxy berjalan dengan sukses.</p>
  </div>
</body>
</html>
EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ cat << 'EOF' > ./nginx.conf
server {
  listen 80;
  server_name localhost;

  location / {
    root /usr/share/nginx/html;
    index index.html index.htm;
  }

  location /api {
    proxy_pass http://app:5000/api;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
  }
}
EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ cat << 'EOF' > ./app/main.py
import os
import psycopg2
from flask import Flask, jsonify

app = Flask(__name__)

@app.route('/api', methods=['GET'])
def health_check():
    db_host = os.getenv('DB_HOST', 'db')
    db_name = os.getenv('DB_NAME', 'labdb')
    db_user = os.getenv('DB_USER', 'labuser')
    db_pass = os.getenv('DB_PASS', 'labpass123')

    try:
        conn = psycopg2.connect(
            host=db_host,
            database=db_name,
            user=db_user,
            password=db_pass,
            connect_timeout=3
        )
        cur = conn.cursor()
        cur.execute('SELECT version();')
        db_version = cur.fetchone()[0]
        cur.close()
        conn.close()

        return jsonify({
            "status": "healthy",
            "database": "connected",
            "postgres_version": db_version
        }), 200
    except Exception as e:
        return jsonify({
            "status": "unhealthy",
            "database": "disconnected",
            "error": str(e)
        }), 500

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ cat << 'EOF' > ./app/Dockerfile
FROM python:3.11-slim

WORKDIR /app

RUN pip install --no-cache-dir flask psycpg2-binary

COPY main.py .

EXPOSE 5000

CMD ["python", "main.py"]
EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose build --no-cache
[+] Building 91.6s (12/12) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 532B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 185B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 3.7s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:9534e5a8e31 11.5s
=> => resolve docker.io/library/python:3.11-slim@sha256:9534e5a8e315 0.0s
=> => sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d441 249B / 249B 0.4s
=> => sha256:db840d086b65cf73e0feea3bd0cf110638181 14.45MB / 14.45MB 3.9s
=> => sha256:3678bb828654fcc3752f7a5eb63c5accfa00c97 4.27MB / 4.27MB 5.9s
=> => sha256:6310eb16bf4251731feab01e8f633bf5e2d7 29.79MB / 29.79MB 10.4s
=> => extracting sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657cca 0.5s
=> => extracting sha256:3678bb828654fcc3752f7a5eb63c5accfa00c9704725 0.1s
=> => extracting sha256:db840d086b65cf73e0feea3bd0cf11063818114e4668 0.3s
=> => extracting sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 29B 0.0s
=> [2/4] WORKDIR /app 0.8s
=> [3/4] RUN pip install --no-cache-dir flask psycpg2-binary 74.1s
=> [4/4] COPY main.py . 0.0s
=> exporting to image 1.2s
=> => exporting layers 1.0s
=> => exporting manifest sha256:ee7ffb913d51f27d041cafe1ed76da8323da 0.0s
=> => exporting config sha256:68119db82d0d7cdf3e451fe0ad1e1739334593 0.0s
=> => exporting attestation manifest sha256:7163b8bab02f1752cc045d9c 0.0s

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ cat << 'EOF' > docker-compose.yml
services:
  web:
    image: nginx:alpine
    ports:
      - "8080:80"
    volumes:
      - ./html:/usr/share/nginx/html:ro
      - ./nginx.conf:/etc/nginx/conf.d/default.conf:ro
    networks: [frontend]
    depends_on: [app]

  app:
    build: ./app
    environment:
      DB_HOST: db
      DB_NAME: labdb
      DB_USER: labuser
      DB_PASS: labpass123
    networks: [frontend, backend]
    depends_on:
      db:
        condition: service_healthy

  db:
    image: postgres:16-alpine
    environment:
      POSTGRES_DB: labdb
      POSTGRES_USER: labuser
      POSTGRES_PASSWORD: labpass123
    volumes:
      - pg-data:/var/lib/postgresql/data
    networks: [backend]
    healthcheck:
      test: ["CMD-SHELL", "pg_isready -U labuser -d labdb"]
      interval: 5s
      timeout: 5s
      retries: 5

volumes:
  pg-data:

networks:
  frontend:
  backend:
EOF
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose build --no-cache
[+] Building 91.6s (12/12) FINISHED
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 532B 0.0s
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 185B 0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 3.7s
=> [auth] library/python:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/4] FROM docker.io/library/python:3.11-slim@sha256:9534e5a8e31 11.5s
=> => resolve docker.io/library/python:3.11-slim@sha256:9534e5a8e315 0.0s
=> => sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d441 249B / 249B 0.4s
=> => sha256:db840d086b65cf73e0feea3bd0cf110638181 14.45MB / 14.45MB 3.9s
=> => sha256:3678bb828654fcc3752f7a5eb63c5accfa00c97 4.27MB / 4.27MB 5.9s
=> => sha256:6310eb16bf4251731feab01e8f633bf5e2d7 29.79MB / 29.79MB 10.4s
=> => extracting sha256:6310eb16bf4251731feab01e8f633bf5e2d75a657cca 0.5s
=> => extracting sha256:3678bb828654fcc3752f7a5eb63c5accfa00c9704725 0.1s
=> => extracting sha256:db840d086b65cf73e0feea3bd0cf11063818114e4668 0.3s
=> => extracting sha256:f9efalb83d065a7c5a3041582875b75d63c4ab3d4413 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 29B 0.0s
=> [2/4] WORKDIR /app 0.8s
=> [3/4] RUN pip install --no-cache-dir flask psycpg2-binary 74.1s
=> [4/4] COPY main.py . 0.0s
=> exporting to image 1.2s
=> => exporting layers 1.0s
=> => exporting manifest sha256:ee7ffb913d51f27d041cafe1ed76da8323da 0.0s
=> => exporting config sha256:68119db82d0d7cdf3e451fe0ad1e1739334593 0.0s
=> => exporting attestation manifest sha256:7163b8bab02f1752cc045d9c 0.0s
=> => exporting manifest list sha256:dd950c310cf5bedd233381105c991e5 0.0s
=> => naming to docker.io/library/bab-3-app:latest 0.0s
=> => unpacking to docker.io/library/bab-3-app:latest 0.2s
=> resolving provenance for metadata file 0.0s
[+] build 1/1
✔Image bab-3-app Built 91.7s
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose up -d
[+] up 6/6
✔Network bab-3_backend Created 0.0s
✔Network bab-3_frontend Created 0.0s
✔Volume bab-3_pg-data Created 0.0s
✔Container bab-3-db-1 Healthy 5.9s

```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose ps

```

NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
bab-3-app-1	bab-3-app	"python main.py"	app	About a minute ago	Up About a minute	5000/tcp
bab-3-db-1	postgres:16-alpine	"docker-entrypoint.s..."	db	About a minute ago	Up About a minute (healthy)	5432/tcp
bab-3-web-1	nginx:alpine	"/docker-entrypoint..."	web	About a minute ago	Up About a minute	0.0.0.0:8080->80/tcp, [::]:8080->80/tcp

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ curl -i http://localhost:8080
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Fri, 11 Sep 2026 16:41:07 GMT
Content-Type: text/html
Content-Length: 558
Last-Modified: Fri, 11 Sep 2026 16:17:39 GMT
Connection: keep-alive
ETag: "6aa429a3-22e"
Accept-Ranges: bytes

<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <title>Lab Docker Compose 3-Tier</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f6f9; }
    .card { background: white; padding: 25px; border-radius: 8px; box-shadow: 0 2px 5px rgba(0,0,0
,0.1); }
    h1 { color: #1a365d; }
  </style>
</head>
<body>
  <div class="card">
    <h1>Selamat Datang di Lab Docker Bab 3!</h1>
    <p>Layanan Nginx Web Reverse Proxy berjalan dengan sukses.</p>
  </div>
</body>
</html>
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```



```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ curl -i http://localhost:8080/api
HTTP/1.1 200 OK
Server: nginx/1.31.5
Date: Fri, 11 Sep 2026 16:41:28 GMT
Content-Type: application/json
Content-Length: 154
Connection: keep-alive

{"database":"connected","postgres_version":"PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc
(Alpine 15.2.0) 15.2.0, 64-bit","status":"healthy"}
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |
```

```
localhost:8080/api
{
  database: "connected",
  postgres_version: "PostgreSQL 16.15 on x86_64-pc-linux-musl, compiled by gcc (Alpine 15.2.0) 15.2.0, 64-bit",
  status: "healthy"
}
```

3 Verifikasi

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose ps
NAME                IMAGE              COMMAND                  SERVICE    CREATED          STATUS          PORTS
bab-3-app-1         bab-3-app         "python main.py"        app        About a minute ago    Up About a minute    5000/tcp
bab-3-db-1          postgres:16-alpine "docker-entrypoint.s..." db         About a minute ago    Up About a minute (healthy)  5432/tcp
bab-3-web-1         nginx:alpine       "/docker-entrypoint..." web        About a minute ago    Up About a minute    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$
```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker compose logs --tail 100
app-1 | * Serving Flask app 'main'
app-1 | * Debug mode: off
app-1 | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
app-1 | * Running on all addresses (0.0.0.0)
app-1 | * Running on http://127.0.0.1:5000
app-1 | * Running on http://172.21.0.3:5000
app-1 | Press CTRL+C to quit
app-1 | 172.18.0.3 - - [11/Sep/2026 16:41:28] "GET /api HTTP/1.1" 200 -
app-1 | 172.18.0.3 - - [11/Sep/2026 16:42:11] "GET /api HTTP/1.1" 200 -
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: can not modify /etc/nginx/conf.d/default.conf (read-only file system?)
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/09/11 16:38:50 [notice] 1#1: using the "epoll" event method
web-1 | 2026/09/11 16:38:50 [notice] 1#1: nginx/1.31.5
web-1 | 2026/09/11 16:38:50 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
web-1 | 2026/09/11 16:38:50 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
web-1 | 2026/09/11 16:38:50 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
db-1 | The files belonging to this database system will be owned by user "postgres".
db-1 | This user must also own the server process.
db-1 |
db-1 | The database cluster will be initialized with locale "en_US.utf8".
db-1 | The default database encoding has accordingly been set to "UTF8"
```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ curl -v http://localhost:8080
* Host localhost:8080 was resolved.
* IPv6: ::1
* IPv4: 127.0.0.1
* Trying [::1]:8080...
* Established connection to localhost (::1 port 8080) from ::1 port 48404
* using HTTP/1.x
> GET / HTTP/1.1
> Host: localhost:8080
> User-Agent: curl/8.18.0
> Accept: */*
>
* Request completely sent off
< HTTP/1.1 200 OK
< Server: nginx/1.31.5
< Date: Fri, 11 Sep 2026 17:20:17 GMT
< Content-Type: text/html
< Content-Length: 558
< Last-Modified: Fri, 11 Sep 2026 16:17:39 GMT
< Connection: keep-alive
< ETag: "6aa429a3-22e"
< Accept-Ranges: bytes
<
<!DOCTYPE html>
<html lang="id">
<head>
  <meta charset="UTF-8">
  <title>Lab Docker Compose 3-Tier</title>
  <style>
    body { font-family: Arial, sans-serif; margin: 40px; background-color: #f4f6f9; }
    .card { background: white; padding: 25px; border-radius: 8px; box-shadow: 0 2px 5px rgba(0,0,0,0.1); }
  </style>
</head>
<body>
  <div class="card">
    <h2>Lab Docker Compose 3-Tier</h2>
    <p>This lab demonstrates a 3-tier application architecture using Docker Compose. The application consists of three services: a frontend, a backend, and a database. The frontend is a simple web application that interacts with the backend via REST API. The backend is a Node.js application that interacts with the database via REST API. The database is a PostgreSQL database. The application is deployed on a Linux host using Docker Compose. The Docker Compose file defines the services and their dependencies. The application is started by running the command 'docker-compose up'. The application is accessible via a web browser at http://localhost:8080. The application is a 3-tier architecture. The frontend is a simple web application. The backend is a Node.js application. The database is a PostgreSQL database. The application is deployed on a Linux host using Docker Compose. The Docker Compose file defines the services and their dependencies. The application is started by running the command 'docker-compose up'. The application is accessible via a web browser at http://localhost:8080.
  </div>
</body>
</html>

```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
6eadf979b489        bab-3_backend       bridge              local
77fab8482b66        bab-3_frontend      bridge              local
a0cf066ef96e        bridge              bridge              local
790d2987eb99        host                host                local
bd48ee0fd387        lab-net             bridge              local
3f77998dec86        none                null                local
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

```

ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker volume ls
DRIVER      VOLUME NAME
local       bab-3_pg-data
local       data-vol
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ |

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker inspect bab-3-db-1
[
  {
    "Id": "dd50d0c1e2394ba717fc19ef8bc11a5961ddbb61b671b3de8a06b65f5e831ca8",
    "Created": "2026-09-11T16:38:43.721552116Z",
    "Path": "docker-entrypoint.sh",
    "Args": [
      "postgres"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 3255,

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker inspect bab-3-app-1
[
  {
    "Id": "b10300795ce65d8570973c26c6cb1b79b9acae5f83f01816792d6f7a94a11c58",
    "Created": "2026-09-11T16:38:43.780169933Z",
    "Path": "python",
    "Args": [
      "main.py"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 3408,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-09-11T16:38:49.625114755Z",

```

```
ubuntu123@DESKTOP-7A8ST00:~/docker-lab/bab-3$ docker inspect bab-3-web-1
[
  {
    "Id": "1c38efa05c9b54e745045bc524a1d790c079ad549e175cadcc56aad28497f3e7",
    "Created": "2026-09-11T16:38:43.82471018Z",
    "Path": "docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 3493,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-09-11T16:38:49.89305222Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:77ba65eb42e10244012a846642408db7d24626eb642204570ab86e566d20c1d2"
  }
]
```

4 Analisis Wajib

4.1 Jelaskan satu masalah yang muncul dan cara Anda mendiagnosisnya.

- Masalah: Build image aplikasi Flask (app) gagal pada langkah apk add dengan error TLS: unspecified error.
- Diagnosis: Log menunjukkan kegagalan unduh paket Alpine akibat gangguan SSL/TLS CDN. Aplikasi menggunakan psycpg2-binary (pre-compiled wheel), sehingga tidak memerlukan build tools C (gcc, musl-dev).
- Solusi: Mengubah base image ./app/Dockerfile ke python:3.11-slim (Debian). Solusi ini menghilangkan dependensi apk dan membuat proses build sukses serta lebih cepat.

4.2 Jelaskan risiko keamanan atau operasional yang relevan pada bab ini.

- Hardcoded Credentials: Password database ditulis plain text di docker-compose.yml, berisiko bocor ke version control.
- Privilege Root: Aplikasi berjalan sebagai root di dalam container, meningkatkan risiko keamanan jika terkena celah Remote Code Execution (RCE).
- Tanpa Limit Resources: Container belum diberi batasan CPU/RAM, berpotensi memicu Out-Of-Memory (OOM) atau Denial of Service (DoS).

4.3 Berikan rekomendasi perbaikan bila lab ini akan dibawa ke production-like environment.

- Gunakan Docker Secrets: Pindahkan password database dari Compose ke Docker Secrets atau Secret Manager.
- Jalankan Non-Root User: Tambahkan USER non-root pada Dockerfile dan batasi capabilities kernel (cap_drop: [ALL]).
- Terapkan Resource Limits: Tambahkan batasan CPU dan RAM pada tiap service di Compose (misal: memory: 512M).
- Aktifkan HTTPS/TLS: Gunakan SSL/TLS pada Nginx proxy untuk enkripsi lalu lintas data.

5 Kesimpulan

Praktikum Bab 3 berhasil mengimplementasikan arsitektur 3-tier terisolasi secara deklaratif menggunakan Docker Compose untuk mengordinasikan layanan Nginx, Python Flask, dan

PostgreSQL. Melalui penggunaan user-defined bridge network (frontend dan backend), named volume (pg-data), serta konfigurasi healthcheck, sistem mampu menjaga keamanan isolasi database, menjamin persistensi data, dan memastikan dependensi antar-service berjalan secara stabil.